Álvaro TONA RIBAS CRUZ

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Summary — Recent Master's graduate from Institut Polytechnique de Paris seeking a PhD position starting in September 2027 or later. My research interests lie in wireless communications, cellular networks, radio mapping and machine learning.

Education

Institut Polytechnique de Paris — **Sep 2025 – Sep 2026**
M.Sc. Electrical Engineering, concentration in Radio Systems (with Highest Honors) — *Palaiseau, France*

Télécom SudParis — **Sep 2024 – Sep 2026**
"Diplôme d'ingénieur", concentration in Electrical and Optical Engineering (with High Honors) — *Évry, France*

Universidade Estadual de Campinas (Unicamp) — **Mar 2021 – Jul 2027**
B.Sc. Electrical Engineering, interrupted from August 2024 to December 2026 for double degree program — *Campinas, Brazil*

Experience

Chaire C2M (LTCI/Télécom Paris) — **Oct 2026 – Jan 2027**
Winter Research Intern — *Palaiseau, France*
- Main project: Prediction of cellular network metrics using Graph Neural Networks (GNNs).
- Side project: Extraction of physical priors from radio-map images for propagation modeling on the lunar surface.

Orange Research — **Mar 2026 – Sep 2026**
Research Intern — *Châtillon, France*
- Summarized a model to calculate KPIs of wireless networks using stochastic geometry and queueing theory.
- Proposed physics-informed machine learning algorithms to predict relevant KPIs in Orange's network.
- Improved SOTA results in simulations and demonstrated prediction limits for current 4G/5G traffic with real data.

Hyperfrequency Lab (EPh/Télécom SudParis) — **Jul 2025 – Sep 2025**
Summer Research Intern — *Palaiseau, France*
- Proposed a method to measure vital signs using a Novelda UWB proximity sensor.
- Simulated a multipath receiver signal-processing chain, including an ADC/DAC, using MATLAB.
- Developed an algorithm to isolate breathing-pattern signals from impulse-radio data and then identify sleep apnea.

Department of Control and Automation (FEEC/Unicamp) — **Mar 2024 – Jul 2024**
Teaching Assistant in "Linear Systems Analysis" — *Campinas, Brazil*
- Created comprehensive review material, including 35 detailed problems & solutions, for 50+ students.
- Main topics: Laplace/Z transforms, linearization, frequency response, state equations, controllability.

Department of Applied Physics (IFGW/Unicamp) — **Apr 2021 – Feb 2023**
Student Researcher for a Scientific Tournament — *Campinas, Brazil*
- Member of Brazil's top competition team for the International Physicists' Tournament (IPT).
- Used Python and Mathematica extensively for numerical simulations in fluid mechanics and electromagnetism.
- Reviewed and discussed scientific papers, built low-cost experimental setups, and gave technical presentations.

Selected Awards

BRAFITEC Scholarship — **2024**
- The most prestigious merit-based scholarship the Brazilian government offers for studies in France. Full ride (∼ 25,000 €). Selected as the third-best candidate from Unicamp across all engineering majors accepted to the double degree program.

2nd Place – XIV International Physicists' Tournament (IPT) — **2022**
- Highest average (8.33), second-best presentation (9.00), and fifth-best review (8.75) out of 86 participants. As a team, we had the best result in Brazil's history in the competition, outperforming universities such as ETHZ/EPFL, UCL, DTU, Polimi.

FEEC Excellence Grant — **2022**
- Exceptional travel grant (∼ 600 €) to represent FEEC/Unicamp at the IPT in Bucaramanga, Colombia.

Silver Honor – International Youth Math Challenge (IYMC) — **2020**
- Scored 19/30 in the final round, placing among the world's top 7% across three rounds while in high school.

Skills

Communications: Information theory, antenna design, MIMO, instrumentation, waveguides, radar, satellites and UAVs, GNSS.
Coding: Python (PyTorch, Keras/TensorFlow, scikit-learn, NumPy, SymPy, SciPy), MATLAB, Mathematica, Arduino, C/C++.
Software: Sionna (RT, PHY), Blender, Advanced Design System (ADS), Altera Quartus II, Git/GitHub/GitLab, LaTeX.
Languages: Portuguese (native), English (C2), French (C2), Spanish (C1), Mandarin Chinese (HSK1).

Publications

[P1] S. Li*, **Á. Ribas***, S. Wang, J. Wiart, "LESMa: Neural Network Architecture for Lunar Surface Radio Mapping," *The First Lunar Pathloss Radio Map Prediction Challenge at ICASSP 2027*, (X).

[W1] **Á. Ribas**, M. K. Karray, "Hybrid Models for Predicting Cellular QoS Using Physics-Informed Machine Learning: From Simulations to Field Measurements," *IEEE Workshop on Communication Networks and Power Systems (WCNPS)*, 2026.

[W2] **Á. Ribas**, Q. Sun, S. Wang, J. Wiart, "Deep Learning-Based Uplink Transmit Power Estimation in 4G/5G Networks," *Journées Scientifiques d'URSI France*, 2026.

[P] Preprint [C] Conference [W] Workshop // * Equal contribution // (X): Not yet submitted

Projects and Theses

[T1] **Álvaro Ribas**, "Prediction of Quality of Service Metrics in Wireless Networks With Physics-Informed Machine Learning," *Master's thesis*, 2026.

[R1] **Á. Ribas***, Z. Luo*, "Indoor Positioning System Using Ultra-Wideband (UWB) Signals," *Research and Development Projects (PRO4501) course final project at Télécom SudParis*, 2025.

[R2] A. C. Andrade*, **Á. Ribas***, "Channel Estimation via Residual Networks for Intelligent Surface-Aided Communication," *Machine Learning (IA048) course final project at FEEC/Unicamp*, 2024.

[R] Project [T] Thesis // * Equal contribution

Reviews

Vehicular Technology Conference (VTC) **2026**

Oral presentations

Meeting of the "Réseau Microondes Île-de-France" (AREMIF) **Nov 2026**
AI architectures for uplink transmission assessment *Paris, France*

Outreach

International Young Physicists' Tournament (IYPT) Brasil **2023, 2024**
Student Jury Member *São José dos Campos, Brazil*

- Evaluated presentations by high school students on various physics topics, such as optics, mechanics and electromagnetism.
- Gave students feedback based on my experience in scientific competitions at the national and international levels.